MCO-1D ORNITHOPTER

Bill of Materials — Revision 17 — July 2026

Full BOM reflecting 8 × ODrive S1 architecture (one board per motor), complete S1 health monitoring across pre-flight / in-flight / post-flight, wing morphing programme H-33 through H-36, pitot airspeed requirement, and all commissioning tooling.

Aircraft	MCO-1D metachronal cascade ornithopter
MTOW	3.448 kg (33.8 N) — measure actual after build
Wings	8 — four tandem pairs W1 to W4
Motors	8 × GoolRC 5010-14 360 KV — one per wing
Motor controllers	8 × ODrive S1 — ONE BOARD PER MOTOR
Encoder	On-board absolute magnetic (1/4" magnet on shaft)
Brake resistors	8 × 2Ω 50W — one per S1, included, cannot share
Health monitoring	Real S1 FET + motor temps, current, rope wear, errors
Pitot sensor	Pixhawk MS4525DO — required for H-33/34/35/36
Wave Library	48,097 lines — 33 hypotheses, 180 variants
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1. AIRFRAME AND STRUCTURE

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	set	Main airframe / fuselage	Mounts 8 sector wheels, 8 motors, 16 servos, 8 ODrive S1 boards, Pi 4, all sensors, pitot tube. Rigid under full-thrust vibration.	Custom
8	pcs	Wing panels — S1223 aerofoil, 165 mm chord	W1=772, W2=749, W3=735, W4=712 mm semi-span (measure actual). Root chord 230-250 mm at bolt circle (tapered extension). Winglets: 75-80° cant, 35° aft sweep, ~140 mm height. Turbulator tape: upper 20-35%, lower 72-75%. Covert flaps: upper 43% and 67% chord. Sweep servo must reach 85-95° aft for H-33 State C.	Custom
8	pcs	Sector wheels — 100° mechanical arc	One per wing. Software limit 96°.	Custom
8	sets	Rope drives — one per wing, independent	No shared rope between port and stbd. Current asymmetry = early rope wear signal.	Cord
1	set	Pitot tube mounting hardware	Forward of all wings in clean airflow. Required: H-33/34/35/36 all use pitot airspeed (V_ms = msg.airspeed). GPS ground speed never used for aerodynamic decisions.	Custom

2. MOTORS — 8 INDEPENDENT

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
8	pcs	GoolRC 5010-14 BLDC 360 KV, 12N14P, 7 pole pairs	One per wing. Verify 5010-14. 7 pole pairs — enter exactly in S1 setup.	~$25
8	pcs	Output drum — 7:1 reduction	7 motor turns = 1 sector wheel turn.	Custom
8	sets	Mount hardware + vibration isolation	Loctite 243 all fasteners.	Hardware

3. ODRIVE S1 MOTOR CONTROLLERS — 8 BOARDS × 1 MOTOR EACH

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
8	pcs	ODrive S1 motor controller	ONE BOARD PER MOTOR. wing_idx == node_id (1:1, no axis0/axis1). Node 0=W1p, 1=W1s, 2=W2p, 3=W2s, 4=W3p, 5=W3s, 6=W4p, 7=W4s. vel_limit=150 rad/s. pole_pairs=7. VELOCITY mode. On-board absolute encoder. CAN-FD capable at 12 Mbps. Configure: python3 mco_odrive_setup.py	~$130

8	pcs	1/4" encoder magnets (6.35 mm diametrically magnetised)	INCLUDED with each S1. Press onto motor shaft centre, flush with back face. Gap to encoder chip: 1-2 mm. Verify: python3 mco_odrive_setup.py --check-magnet (status 1=OK, 0=missing, 2=too close, 3=too far).	Incl.
8	pcs	Brake resistors — 2Ω 50W	INCLUDED with each S1. One per board. CANNOT be shared between boards. Each S1 has its own DC bus — requires its own brake resistor.	Incl.
1	pcs	MCP2515/2518FD SPI-to-CAN adapter	SPI0 CE0. 8 MHz osc. Int GPIO25. Daisy: Pi → node0 → 1 → 2 → 3 → 4 → 5 → 6 → 7.	~$5
2	pcs	120Ω CAN termination resistors	One at Pi end, one at node 7 end.	<$1
2+	m	CAN twisted pair cable	Daisy-chain all 8 S1 boards.	Per m

4. PITCH SERVOS — ANGLE OF ATTACK — 8 WINGS

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
8	pcs	Digital servo with feedback pot — pitch (AoA)	Feedback wire (3rd pin) required. 333 Hz PWM. 7.4 V supply. GPIO: W1p=18,W1s=23, W2p=24,W2s=25,W3p=12,W3s=16,W4p=20, W4s=21. Feedback → 1kΩ → MCP3208 #1 CE1. CH0-CH7.	~$15
8	pcs	1kΩ resistors pitch feedback	Series protection per line.	<$1

5. SWEEP SERVOS — FORE-AFT / DYNAMIC SWEEP / STOW — 8 WINGS

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
8	pcs	High-torque digital servo with feedback pot — sweep	Feedback wire required. 25 kg·cm+. 7.4 V. Must reach 85-95° aft for H-33 State C stow. Must execute ±20-40° in <200 ms for H-34 dynamic sweep. GPIO: W1p=4,W1s=17,W2p=27,W2s=22,W3p=5,W3s=6,W4p=13,W4s=19. Feedback → 1kΩ → MCP3208 #2 CE0. CH0-CH7.	~$12
8	pcs	1kΩ resistors sweep feedback		<$1

6. ADC CHIPS — PER-WING SENSING

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
2	pcs	MCP3208-CI/P 8-ch 12-bit SPI ADC	#1 PITCH: SPI1 CE1 (GPIO7). #2 SWEEP: SPI1 CE0 (GPIO8). Both share SPI1: CLK=GPIO11, MOSI=GPIO10, MISO=GPIO9. VREF=VDD=3.3V. /boot/config.txt: dtoverlay=spi1-2cs	~$3

7. FLIGHT COMPUTER

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	pcs	Raspberry Pi 4 Model B — 4 GB RAM	Mean tick ~13 ms at 50 Hz. Vibration isolate.	~$55
1	pcs	MicroSD 64 GB high-endurance	~50 MB/hr at 50 Hz.	~$12
1	pcs	Pi heatsink + active cooling case		~$10

8. SENSORS

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	pcs	MPU-6050 IMU — I2C 0x68	I2C1 SDA=GPIO2 SCL=GPIO3.	~$3

1	pcs	BMP280 barometer — I2C 0x76	Same I2C1 bus.	~$3
1	pcs	TF-Luna lidar — 50 Hz	Facing down, clear of wing arc.	~$20
1	pcs	DS18B20 OAT sensor	GPIO4 One-Wire. 4.7kΩ pull-up. Ambient reference for pre-flight temperature check.	~$3
1	pcs	Pitot tube — Pixhawk MS4525DO	REQUIRED for H-33/34/35/36. V_ms = msg.airspeed (pitot only, not GPS ground speed). Mount forward of all wings in clean airflow.	~$20
1	pcs	HX711 + 50N load cell — bench only	TC-01 and TC-02. Not flown. HX711: DT=GPIO26, SCK=GPIO7. Calibrate with 1 kg mass: mco_loadcell.py --calibrate	~$12

9. RADIO AND NAVIGATION

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	set	RC transmitter + SBUS receiver — 4 channels	CH1=Roll, CH2=Pitch, CH3=Lever, CH4=Yaw. Verify: mco_rc_verify.py. SBUS → GPIO15.	Per system
1	pcs	Pixhawk — sensor bridge only	GPS + compass + IMU + PITOT AIRSPEED → Pi via MAVLink. TELEM2 → Pi UART 921,600 baud.	~$40
1	pcs	GPS module for Pixhawk	Mount away from motors.	~$20
1	pcs	Touchscreen 3.5 inch+	mco_panel.py: 4 buttons, 6 margin bars. Motor health alerts feed into margin bar. CRITICAL motor alert overrides aerodynamic margin display.	~$25

10. POWER SYSTEM

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	pcs	Main LiPo — 4S 4000 mAh+ 30C	8 × S1 + 8 motors. LOW 30%, CRITICAL 20%.	Per spec
1	pcs	Servo LiPo — 2S 1500 mAh 7.4V	DEDICATED for all 16 servos.	Per spec
1	pcs	BEC 5V/3A isolated	Pi supply. ISOLATED from motor rail.	~$8
1	pcs	30A fuse + holder	Main battery lead.	~$3
1	pcs	5A fuse + holder	Pi BEC.	~$2

11. COMMISSIONING EQUIPMENT

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	set	Static test post + load cell fixture	Aircraft bolts to post. Load cell inline. Must withstand >40N. Used for TC-01 (bench) and TC-02 (captive hover).	Custom
1	pcs	Load cell — 50N+ compression	Reads zero at hover frequency. mco_loadcell.py via HX711.	~$15
1	pcs	Calibration mass — 1.000 kg known	mco_loadcell.py --calibrate before TC-01.	~$10
1	pcs	Truck / flatbed — TC-03	Forward airspeed 3-15 m/s. GS shows target speed.	Borrowable
1	pcs	IR thermometer	Independent motor temp verification at TC-01. S1 also reports real temps over CAN from TC-01 onwards.	~$20
1	pcs	Laptop — Python 3.8+, WiFi	Ground station.	Per availability
1	pcs	Tape measure 2 m	mco_build_measurements.py.	~$5

1	pcs	Kitchen scale 5 kg / 1 g	MTOW measurement.	~$15
1	pcs	Protractor / digital angle gauge	Sector arc, sweep range, stow angle measurement.	~$10
1	pcs	ST-Link V2 programmer	S1 board recovery if firmware bricked. Not normally needed.	~$10
1	set	Field safety kit	LiPo extinguisher. First aid. Safety glasses. 30 m clear zone.	Per site

12. CONSUMABLES

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	roll	Loctite 243 blue thread lock	All fasteners.	Per roll
8	pcs	Spare 1/4" encoder magnets (6.35 mm)	Same spec as S1-included. Replacements if shaft magnet comes loose.	~$10
2	pcs	Spare rope drive sections		Per length
1	roll	Zigzag turbulator tape 0.10-0.15 mm	Upper surface 20-35% chord. Lower 72-75% chord.	~$15

13. SOFTWARE (all open source)

Qty	Unit	Item — Full Specification	Notes / Wiring	~Cost
1	file	mco_wave_library.py — 48,097 lines	33 hypotheses H-01 to H-36. 180 variants. 148 GS commands. H-33/34/35/36 morphing ticks wired into main tick wrapper. S1 FET temps feed ThermalMonitor directly. Dynamic vel_limit from real thermal data.	No cost
1	file	mco_odrive_interface.py — 447 lines	8 × S1 CAN. wing_idx == node_id. read_temperatures(), read_all_states(), set_velocity_limit(), clear_errors(), estop(), set_initial_phases().	No cost
1	file	mco_odrive_setup.py — 282 lines	Configures 8 S1 boards. --verify --calibrate --check-magnet.	No cost
1	file	mco_preflight_checks.py — 1,256 lines	22 check functions. PRE-FLIGHT (19 checks): S1 heartbeats, encoder magnets, temperatures, axis errors, current baseline. IN-FLIGHT: S1InFlightMonitor at 50 Hz — FET temp (warn 70°C crit 90°C), motor temp (warn 60°C crit 75°C), current (warn 18A crit 22A), rope wear (>20% rise), P/S asymmetry (>25%), axis errors. POST-FLIGHT: check_s1_post_flight() — peak temps, rope trend, cooling state, errors, next-flight GO/NO-GO.	No cost
1	file	mco_main.py — 483 lines	S1 temps at 5 Hz; errors at 1 Hz. Dynamic vel_limit per-wing from thermal data. Auto clear_errors per tick. Post-flight S1 check at shutdown. Motor health telemetry to GS and panel.	No cost
1	file	mco_panel.py — 506 lines	Motor health alerts on margin bar. CRITICAL overrides aerodynamic margin display.	No cost
1	file	mco_loadcell.py — 527 lines	HX711. Calibrate/tare/sweep/hover.	No cost
1	file	mco_build_measurements.py — 328 lines	15 measurements, patches WL.	No cost
1	file	mco_preflight.py — 782 lines	Three-tier preflight original.	No cost
1	file	mco_rc_verify.py — 324 lines	RC mapping + lever interlock.	No cost
1	file	mco_vibration_check.py — 367 lines	Vibration sweep. SPI stability.	No cost
1	file	mco_test_cards.py — 1,665 lines	TC-01 through TC-04.	No cost
1	file	mco_log_analysis.py — 462 lines	Post-flight log analysis.	No cost
1	file	mco_ground_station.py — 660 lines	motor_health, motor_peaks, morphing_speed_report, loiter_endurance_start, h33-h36 status, tc_results.	No cost

9+	files	Supporting files	mco_gs_server.py, mco_logger.py, mco_mavlink.py, mco_servo_interface.py, mco_hypothesis_tool.py, mco_sync.py, mco_can_setup.sh, mco_install.sh, mco_performance_model.py, mco_taxi_park.py.	No cost

COMPLETE HARDWARE COUNT — REVISION 17

Wing motors	8 — one per wing, all independent
ODrive S1 boards	8 — ONE PER MOTOR (node_id = wing_idx, 1:1)
Encoder magnets	8 — included with S1, press onto motor shaft
Brake resistors	8 × 2Ω 50W — included with S1, NOT shared
Pitch servos	8 — feedback pot required
Sweep servos	8 — feedback pot, must reach 85-95° aft
Total servos	16
Pitch ADC	1 × MCP3208 SPI1 CE1 (GPIO7)
Sweep ADC	1 × MCP3208 SPI1 CE0 (GPIO8)
1kΩ resistors	16 (one per servo feedback line)
CAN bus	8 S1 nodes daisy-chained, 120Ω at each end
Pitot tube	1 — REQUIRED for H-33/34/35/36 morphing
Load cell	1 × 50N + HX711 — bench only (TC-01/02)
Flight computer	Raspberry Pi 4, 4 GB RAM, 64 GB SD
Pre-flight checks	19 (inc. 5 S1-specific)
In-flight monitors	6 per wing: FET temp, motor temp, current, rope, asym, errors
Post-flight checks	5 (peak temps, rope trend, cooling, errors, readiness)
Wave Library	48,097 lines Python
Hypotheses / variants	33 / 180
GS commands	148
Discovery modes	12
Total Python code	65,336 lines across 31 files
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■ WARNING: 8 × ODrive S1 = 8 boards × 1 motor each. Do NOT share brake resistors between boards — each has its own DC bus.

■ WARNING: All morphing decisions use PITOT AIRSPEED (V_ms = msg.airspeed from Pixhawk). GPS ground speed is never used for any aerodynamic calculation.

● CAUTION: S1 encoder magnets are included with each board. Press onto motor shaft, 1-2 mm gap to encoder chip. Run mco_odrive_setup.py --check-magnet to verify field status.

✓ Motor health monitoring uses real S1 FET and motor temperatures — not estimates. Pre-flight checks, in-flight monitoring, and post-flight assessment all use actual measured values from the S1 boards over CAN. Health alerts appear on the panel margin bar and ground station.